Quantum Mechanics II — Exercise Sheet 13

Angular momentum and radial states

Variant: Questions only

Exercise 1 (A cubic harmonic decomposition). In three dimensions let $p(x) = (x_1 + \mathrm{i}x_2)^3$. (i) Show directly that $\Delta p = 0$ and find $L_3 p$. (ii) Identify its orbital spin and weight on the unit sphere. (iii) Then decompose x_3^3 into a homogeneous harmonic polynomial of degree three and r^2 times a harmonic polynomial of degree one.

Exercise 2 (The first radial excitation of the oscillator). Let $s = \sqrt{\hbar/(m\nu)}$ and $l \geq 0$ be an integer. For the radial oscillator expression

$$H_l = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{\hbar^2 l(l+1)}{2mr^2} + \frac{m\nu^2 r^2}{2},$$

first verify the energy of $u_0(r) = r^{l+1}e^{-r^2/(2s^2)}$. Then find c and E such that $u_1(r) = r^{l+1}(c - r^2/s^2)e^{-r^2/(2s^2)}$ satisfies $H_l u_1 = E u_1$ for $r > 0$. These are radial expressions; no new self-adjoint realization on the half-line is to be proved.

Exercise 3 (A second radial excitation). Use the radial oscillator expression from Exercise 2, now with $l = 0$. Put $z = r^2/s^2$ and seek a solution

$$u(r) = r\left(z^2 + Az + B\right)e^{-z/2}, \qquad r > 0.$$

(i) Find A, B and the energy E for which $H_0 u = Eu$ as a radial differential identity. (ii) Find the two positive radial zeros and identify the total oscillator excitation N from $E = \hbar\nu(N + 3/2)$. Here H_0 denotes the $l = 0$ radial expression of Exercise 2, not the free Hamiltonian on the line.

Exercise 4 (Angular probabilities without a classical orbit). On the unit sphere consider $Y(\theta, \varphi) = C \sin\theta \cos\theta\, e^{\mathrm{i}\varphi}$, with $C > 0$. (i) Normalize it with surface measure $\sin\theta\, \mathrm{d}\theta\, \mathrm{d}\varphi$. (ii) Find the probability of the northern hemisphere and $\langle \cos^2\theta \rangle$. (iii) Identify l and μ using the harmonic polynomial $x_3(x_1 + \mathrm{i}x_2)$.